VIDYAPEETH

BATCH CODE: 12-AN-181-MA-2026

SUBJECT NAME: PHYSICS

CHAPTER NAME:
UNITS & MEASUREMENT

Lecture No. 02

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Today's Goal

Subtopic .. Fundamentals of PHYSICS

Subtopic .. PHYSICAL QUANTITIES.

Subtopic ..

PHYSICS :- Scientific study of Nature.

CLASSIFICATION OF PHYSICS

NEWTONIAN PHYSICS
[classical physics]

* MACROSCOPIC STUDY

Eg :- $m_{UNIVERSE} \approx 10^{55}$ kg

$m_{Pen} \simeq 10$ g

QUANTUM PHYSICS

MICRCOSCOPIC STUDY

Eg :- $m_e = 9.1 \times 10^{-31}$ kg

XII – • Atom & Nucli;
• Semiconductors.

Newtonian

wave
- XI Wave motion
- XII wave optics

XI (1) MECHANICS - Study of Motion & Force.

XII (2) OPTICS
- GEOMETRICAL OPTICS (Mirr/Lens)
- WAVE OPTICS ... → wave Theory of Light. { Eg:- POLARISATION, INTERFERENCE

XII (3) ELECTROMAGNETISM :- Study of Electrostatics/Electrodynamics Magnetism ....

XI (4) THERMODYNAMICS :- Study of Heat

FUNDAMENTAL FORCES IN NATURE :-

(m, q) :- fund prop

1. GRAVITATIONAL Force. (By Virtue of masses)
   - Range :- ∞
   - WEAKEST FORCE in nature

2. ELECTROMAGNETIC force :- (due to charged particles)
   Range :- ∞

10^{39}

3. STRONG Nuclear force :- (b/w p^+ & neutron)
   (Strongest force in Nature)
   | Very short Range [10^{-15} m]

4. WEAK Nuclear force :- nucleus & e^-
   | Short Range (10^{-16} m)

FUNDAMENTAL LAWS IN NATURE :-

1. Conservation of Energy
2. " " Charge
3. " Linear Momentum
4. ★ " " Angular Momentum

$\theta \rightarrow$ Angular displ.

$\omega \rightarrow \frac{\theta}{t} \rightarrow$ Ang. vel.

✓ Unit & M.

Mathematical Tool

Trigo

Alg.

A.P

G.P

log ★

Calculus < Diff ✓

Inte

Motion in 1D

vector 2D

NLM

WEP

Rot. M

Grav

1. Potential Energy

2. Angular Mom.

Vector

★★★★★

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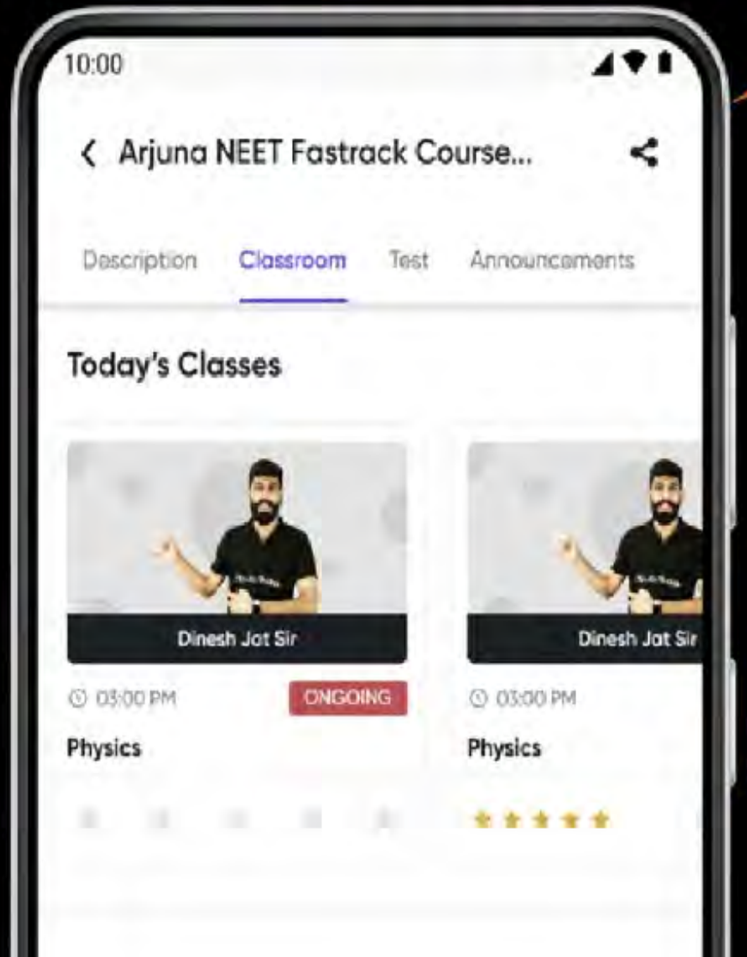
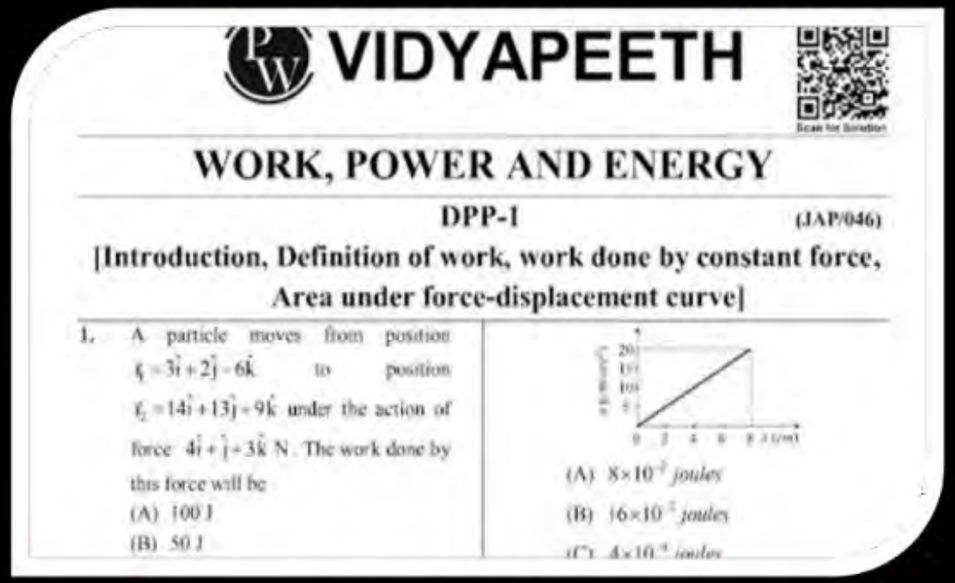


Solve the DPP and check Solution



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